



Exploratory Data Analysis: the diamonds dataset

Exam Proctor faculty pilot · Monday 5 October 2026 · Part B, 11:38 to 12:15 · Old IQAC · 40 marks

Instructions

- Work in **Google Colab**: open `colab.research.google.com` and choose **New notebook**.
- You may also use the documentation at `pandas.pydata.org`, `seaborn.pydata.org` and `matplotlib.org`. Every other website, and every other app on the laptop, is reported.
- Type the code below into the first cell and run it with **Shift + Enter**. Then answer each question in its own cell; add a text cell where a question asks you to explain.
- This is a pilot of the monitoring tool. Nobody is graded; the marks show how much detail is expected.

First cell: load the data

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('diamonds') # 53,940 diamonds, built into seaborn
df.head()
```

Questions

Question 1

2 marks

Load the diamonds dataset and report the number of rows and columns.

Question 2

3 marks

Show the data type of every column. Which columns are categorical?

Question 3

3 marks

How many missing values are there in the whole dataset? How many rows are exact duplicates?

Question 4

4 marks

For `price` and `carat`, report the mean, median, minimum and maximum. Is `price` right-skewed? Justify using the mean and the median.

Question 5**4 marks**

How many diamonds are there of each `cut` ? Which cut is the most common, and what percentage of all diamonds is it?

Question 6**6 marks**

Compute the mean price for each `cut` , sorted from highest to lowest. The best cut (Ideal) has the lowest mean price. Compute the mean `carat` for each cut and use it to explain why.

Question 7**4 marks**

Plot a histogram of `price` . Describe its shape in one sentence.

Question 8**4 marks**

Draw a scatter plot of `carat` (x) against `price` (y). Use transparency or a random sample of 5,000 rows so the points are readable. Describe the relationship.

Question 9**4 marks**

Draw a box plot of `price` for each `color` . Which colour has the highest median price? (D is the best colour and J the worst.)

Question 10**6 marks**

Compute the correlation matrix of the numeric columns. Which variable is most strongly correlated with `price` , and what is the correlation? (Optional: show it as a heatmap.)

Dataset: `diamonds` from the seaborn library: 53,940 diamonds with price, carat, cut, colour, clarity and dimensions. End of question paper.